Problem Chosen	2020 MCM/ICM Summary Sheet	Team Control Number
C		7391856

From Review Signals to Product Decisions

Summary

Sunshine needs to choose product investigations and track new listings using ratings, written reviews and helpfulness. We analyze all 32,024 supplied records, then distinguish observed reviewing, the meaning of feature mentions and the information available at a monitoring cutoff.

Microwaves have a 31.8% one–two-star share, versus 14.6% for dryers and 11.3% for pacifiers. Within the target-title cohorts, reliability words accompany 38.07 and 41.10 percentage points more low-star reviewing inside dryer and microwave families; every one-family deletion keeps both directions positive. A fixed 58-review semantic audit shows why the current unit, previous product, attribute direction and use context must be checked. Low-star reviews can praise light weight or fast drying while describing failure; pacifier words can concern packaging, accessories or other meanings. The pooled material association largely disappears within families. Four reliability-hit strata have conditional simultaneous 95% low-star contrast ranges of [4.55,86.96] and [30.94,75.16] points under a hash-rank randomization model and fixed AI coding; unrestricted compatible ranges cross zero. Unchecked non-hit failures prevent a total-failure conclusion. We route current events to investigation, old-product mentions to monitoring and misrouted material records to recalculation.

For launch tracking, retain the full five-star distribution, ninety-day low-star risk with count and a Beta interval, verified strata, current-unit complaint counts, vote coverage and conditional helpfulness. Fixed-family comparisons reveal that listing mix can mask reputation change. An explicit loss rule distinguishes acceptable own-product risk from deterioration relative to an already poor competitor reference.

The original common-window forecast gives combined log loss 0.323683 versus rating-only 0.324025 over 3,901 future reviews. The small gain's saved family-cluster interval spans zero. Explicit later updates in text and a changed-star example show that a review date does not establish historical field availability. Prefix and blank-marked text challenges give 0.323654 and 0.323684, respectively: the nominal increment changes little, but the scores remain conditional on current snapshots. Begin a timestamped own-product ledger and prospectively validate the simpler rating warning baseline; text primarily routes investigations.

Separating all five star levels in the same family/calendar panel reveals no clear association with later ninety-day review volume. Latest-three-review comparisons retain only identifiable day order; adverse runs are sparse and fitted intervals span zero. Missing purchase and display exposure also prevent causal incitement identification. Sunshine can act on the reliability evidence and monitoring policy while keeping feature preferences, sales effects and rating-campaign claims conditional on the observations they require.

Keywords: review reputation; attribute context; measurement sensitivity; information time; product investigation

Contents

Letter to the Marketing Director **3**

1 Decisions that the supplied reviews can inform **4**

2 Assumptions and their consequences **4**

3 Review populations and helpfulness **5**

3.1 Product and reviewer composition 6

3.2 Attention and endorsement of a review 6

4 From words to design investigations **7**

4.1 Quality descriptors across rating levels 7

4.2 Family-adjusted investigation priorities 7

4.3 A fixed semantic audit and its action implications 9

5 Monitoring reputation on a real calendar **12**

5.1 A dashboard with a decision rule 12

5.2 Time trends and listing mix 12

6 Warning models and the information they require **13**

6.1 A common prediction task. 13

6.2 The timestamp challenge changes the interpretation 15

7 Specific stars, sequences and subsequent reviews **16**

8 A practical launch strategy and its limits **17**

A Reproduction files and fitted parameters **20**

Report on Use of AI **21**

Letter to the Marketing Director

Dear Marketing Director,

Start with a reliability investigation for the new hair dryer and microwave. The supplied competitor reviews show 38.07 and 41.10 percentage points of excess low-star reviewing within product families when reliability words occur. Every one-family deletion preserves these directions. Confirm the current unit and event before engineering inspects relevant returns. Stoppage identifies a symptom, not a failed motor. Four small reliability-hit strata give broad conditional 95% contrast ranges [4.55,86.96] and [30.94,75.16] points under hash-rank sampling and fixed AI coding. Compatible ranges cross zero and unhit failures are unchecked, so the priority is investigation rather than a measured physical defect rate.

The design lesson is to retain useful performance while resolving the specific failure. Two-star dryer reviews can praise fast drying before stoppage; a one-star review can like lightweight construction while condemning failed heat control. Assess handling, heat control and drying performance together. For microwaves, a reported run-without-heat event merits inspection; it alone names no defective component. Monitor prior-machine comparisons separately. For pacifiers, packaging plastic is a packaging observation; remove it from nipple-material counts and recompute the ranking. Verify product scope and acceptance context before selecting material or size. The material association falls to only 0.23 percentage points within families and does not justify one universal material choice.

From the first sale, maintain a timestamped rating/text ledger and a weekly dashboard over the preceding ninety days. Display all five star counts, low-star fraction and interval, verified strata, current-unit issue counts, vote coverage and helpfulness among voted reviews. Management should set an acceptable reviewer-risk tolerance and an investigation budget independently of competitor averages. In the supplied final-window example, five low stars among eleven reviews do not trigger deterioration above a poor microwave reference. Yet an explicitly illustrative 25% tolerance with false-investigation/missed-excess costs 1:4 gives a 97.33% excess probability and favors investigation. Those costs are a policy example for you to replace with actual company values.

Use ratings as the initial warning baseline and text to explain and route the alert. Combined text/ratings improve historical snapshot loss by only 0.000342 nats per review, with a family-cluster interval spanning zero. Removing marked update passages barely changes that small increment, but current text and even stars can have changed after the recorded date. Validate the warning rule against Sunshine's own timestamped future records before relying on its probability scores.

Finally, specific-star and latest-three-review comparisons provide no clear adjusted review-volume association; sparse runs and missing displayed histories limit the answer. Do not budget a campaign on the belief that a string of poor ratings causes more reviewing. Our strongest justified expenditure is a source-verified electrical reliability investigation coupled with transparent own-listing monitoring.

Sincerely,
Team 7391856

1 Decisions that the supplied reviews can inform

Sunshine Company needs three decisions before and after introducing a hair dryer, a microwave oven and a baby pacifier: which design problems deserve investigation, how to monitor each listing's reputation, and which warning signals are worth maintaining. The three supplied files provide 32,024 ratings, written reviews and cumulative votes on reviews [1]. We use the files as evidence about reviewing customers. A sale, an exposed shopper and a physical failure are different observations and are absent from the tables.

Our answer has three connected parts. First, product scope and the meaning of a feature mention determine whether a complaint can be routed to the right engineering team. Second, a calendar dashboard distinguishes a changing dissatisfied tail from changing listing composition. Third, forecasting and review-flow comparisons determine whether more complicated signals add enough information to justify deployment. The strongest action is a reliability investigation for the electrical products; the text is read as evidence for a specific problem before any design change is approved.

Let $r_i \in \{1, 2, 3, 4, 5\}$ be review i's stars, $b_i = \mathbf{1}\{r_i \leq 2\}$ its low-star indicator, and $j(i)$ its supplied `product_parent`. For a set of reviews, n is its size, $k = \sum b_i$, $q = k/n$ and $\bar{r}$ is the mean rating. The denominator of q is reviews. We retain every supplied row in the primary description and give a separate target-title sensitivity for transferring findings to Sunshine's intended products.

The task mapping is direct: Section 3 covers cross-field patterns; Section 4 addresses descriptors and design themes; Section 5 gives tracking measures and time patterns; Section 6 compares text and ratings for future reviewer reputation; and Section 7 examines specific stars and sequences in relation to later review volume. The letter states the decision we recommend most confidently.

2 Assumptions and their consequences

1. **The population is observed reviewers.** Unknown extraction and willingness to review prevent a purchaser defect-rate estimate. The same limitation makes review count a measure of feedback volume rather than sales.

2. **Parent IDs define the supplied comparison unit.** Variants may share a family. Within-family comparisons reduce stable listing differences, while decisions still require identifying the actual unit and use context.

3. **One–two stars define dissatisfied reviewing.** The supplied scale supports this operational definition. We retain the full five-level histogram so that intermediate ratings and different kinds of low-rating histories remain visible.

4. **Words are fallible measurements.** Dictionaries measure mention patterns; a reading protocol records object, attribute, direction and time. The semantic audit is performed by one author AI, with no independent human reference labels. Its coded summaries are conditional measurement scenarios.

5. **Snapshot fields may have been edited.** A review date does not establish when every text passage or current star value became available. Historical forecast scores therefore condition on the supplied snapshot representation. Explicit update-marker censoring challenges this assumption without restoring unavailable historical versions.

6. **The dashboard's prior and losses are policy choices.** Ten pseudo-reviews stabilize sparse estimates; strengths two and thirty challenge that choice. Illustrative investigation costs do not represent Sunshine's accounts.

7. **Calendar/family adjustment supports association.** Purchases, impressions, displayed ratings and changing product quality remain unobserved. No assumption equates an adjusted review-flow coefficient with causal incitement.

3 Review populations and helpfulness

Table 1. Supplied review populations.

Category	Reviews	Families	Mean	Low-star	Voted
Hair dryer	11,470	473	4.116	14.6%	37.7%
Microwave	1,615	55	3.445	31.8%	66.9%
Pacifier	18,939	5432	4.305	11.3%	28.0%

All rows have usable review dates and one–five stars; review IDs are unique within each file. Helpful votes are nonnegative and never exceed total votes. The files share a final recorded date of August 31, 2015, but start at different times. The corresponding means and histograms describe unequal supplied populations rather than an equal market experiment.

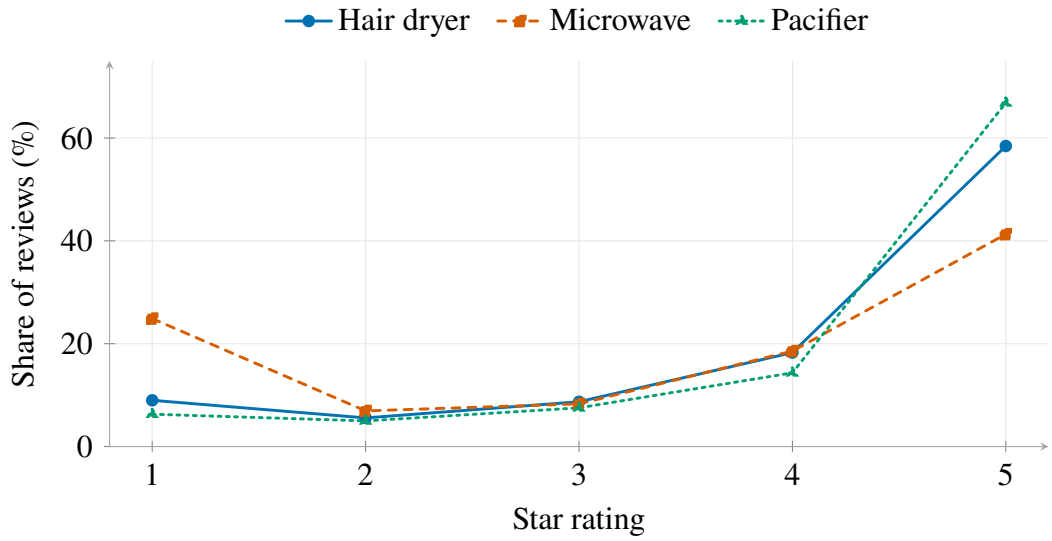


Figure 1. Five-level rating profiles.

Each series uses its full supplied category denominator. Connecting lines guide comparison of discrete star levels.

Microwave reviews have the largest dissatisfied tail, 31.8%, despite 41.3% five-star reviews. Hair dryers and pacifiers have low-star shares of 14.6% and 11.3%. Tracking only a mean would conceal this tail. Sunshine should retain q, n and all five counts together: a reduction in one-star reviews and a shift from four to five stars are different changes even if the mean moves by the same amount.

3.1 Product and reviewer composition

The extracted files include accessories and unrelated products. Our stricter cohort requires positive title expressions (hair dryer/hairdryer/blow dryer/blowdryer; microwave; pacifier/paci/soothie) and excludes a declared category-specific accessory dictionary. Hood dryers remain eligible; holders, replacement parts and explicitly titled clips/cases are separated. This reproducible lexical cohort gives a useful scope challenge, while a diaper bag containing the words "pacifier bag" demonstrates that it still needs product verification.

Table 2. Target-title sensitivity.

Category	Reviews	Families	Mean stars	Low-star
Hair dryer	11,035	123	4.122	14.4%
Microwave	1,467	39	3.337	34.4%
Pacifier	10,475	286	4.419	9.1%

Only 10,475 of 18,939 pacifier-file rows meet the narrow title rule; their low-star fraction is 9.1%. The full file is an extracted review population, and the title cohort is a closer product proxy rather than a verified set of genuine pacifiers. This scope difference is why aggregate baby-product findings cannot become an unqualified pacifier design preference. The target cohort was defined after the original forecast outcomes were inspected and is a secondary sensitivity.

Purchase and Vine flags contain both upper- and lowercase forms. After case normalization, verified microwave reviews have 15.7% low stars versus 65.8% among other microwave reviews; dryer figures are 12.7% versus 25.5%, and pacifier figures 10.3% versus 17.2%. These differences could reflect product mix, timing and reviewing selection. Sunshine should show the two strata and read both, rather than dismissing unverified complaints or assuming verification creates product quality.

3.2 Attention and endorsement of a review

For a review with at least one vote, helpfulness is $u_i = h_i/v_i$. A review with no votes has no measured helpfulness judgment. The aggregate $\sum h_i / \sum v_i$ weights voted reviews by vote count; it answers a different question from the proportion of all reviews that receive attention.

Vote coverage is 37.7% for hair dryers, 66.9% for microwaves and 28.0% for pacifiers. One-star reviews receive votes more often than five-star reviews (67.9% versus 31.7% for dryers; 79.1% versus 56.8% for microwaves; 64.3% versus 21.5% for pacifiers). Yet their helpful-vote fractions are lower. Complaints attract attention without every voter endorsing them.

Review age and text length correlate positively with total votes; length correlations remain positive within age quartiles. Helpful votes thus combine reception, exposure and time. Use

Table 3. Reception and voting exposure.

Category	Helpful/votes	Age–votes ρ	Length–votes ρ
Hair dryer	85.0%	0.31	0.37
Microwave	84.3%	0.45	0.46
Pacifier	73.1%	0.23	0.36

heavily voted complaints to prioritize reading, and display vote coverage and conditional helpfulness separately. Cumulative vote snapshots have no vote timestamp and do not enter the historical warning model.

4 From words to design investigations

4.1 Quality descriptors across rating levels

We tokenize headline plus body after lowercasing and removing HTML. The positive dictionary includes love/loves/loved, excellent, perfect, great, amazing and recommendation forms; the adverse dictionary includes disappointment, terrible/horrible/awful, broken/broke/breaks/defective, stopped and return/refund forms. A preceding-three-token negation heuristic suppresses some positive hits, with explicitly tested positive-intensification exceptions: "cannot recommend enough" and "could not be more perfect". It is a surface measurement, not a universal sentiment parser.

For low/non-low counts (a, b) with a descriptor and (c, d) without it, the corrected odds ratio is

$$O = \frac{(a + 1/2)(d + 1/2)}{(b + 1/2)(c + 1/2)}. \tag{1}$$

The half-count keeps empty cells defined. A count-based 95% log-odds interval uses standard error $\{\sum_{z\in\{a,b,c,d\}}(z + 1/2)^{-1}\}^{1/2}$. It describes pooled lexical association and does not account for family clustering or certify the phrase's meaning.

Love and perfect expressions associate with high ratings, whereas disappointment and broken expressions associate with low ratings in all three categories. Full five-star descriptor counts and all 33 declared comparisons remain in the numerical attachment. These are useful descriptor–rating patterns for question 2(e). A review can nevertheless praise one attribute and condemn another: stars summarize the author's overall evaluation, while a design decision needs the particular attribute's direction.

4.2 Family-adjusted investigation priorities

Topic dictionaries measure reliability, weight and power/speed for dryers; reliability and size/capacity for microwaves; and material, acceptance and related pacifier issues. The topic measurement includes favorable and adverse language. For families with at least three mentioning and three non-mentioning reviews, define

$$D = \frac{\sum_j w_j(q_{j1} - q_{j0})}{\sum_j w_j}, \qquad w_j = \frac{n_{j1}n_{j0}}{n_{j1} + n_{j0}}. \tag{2}$$

Table 4. Descriptor associations with low-star reviewing.

Cat.	Descriptor	n	Low-star	OR	95% interval
H	love	2668	3.9%	0.19	[0.16, 0.23]
H	perfect	617	3.6%	0.21	[0.14, 0.32]
H	disappointed	345	57.1%	8.71	[6.99, 10.85]
H	broken	434	48.6%	6.20	[5.10, 7.55]
M	love	183	13.1%	0.30	[0.19, 0.46]
M	perfect	155	3.2%	0.07	[0.03, 0.16]
M	disappointed	49	65.3%	4.17	[2.31, 7.54]
M	broken	105	82.9%	11.99	[7.17, 20.04]
P	love	6846	2.8%	0.15	[0.13, 0.18]
P	perfect	1453	1.6%	0.12	[0.08, 0.18]
P	disappointed	400	59.8%	13.00	[10.59, 15.96]
P	broken	237	50.6%	8.48	[6.55, 10.98]

Families with large, balanced groups receive more weight; a large family with very few mentioning reviews receives less. This comparison asks whether the lexical signal persists inside supplied families instead of being produced solely by different listing mixes.

Table 5. Design topics in the target cohort.

Cat.	Topic	n	Low-star	Within (pp)	Families
H	reliability	959	51.3%	+38.07	81
H	weight	1932	9.7%	-5.50	102
H	power/speed	3275	8.1%	-8.32	120
M	reliability	255	80.0%	+41.10	33
M	size/capacity	480	17.1%	-17.27	24
P	material	548	14.8%	+0.23	62
P	acceptance	1465	9.8%	-0.14	134

Reliability mentions accompany low-star shares of 51.3% for target dryers and 80.0% for target microwaves, with within-family excesses of 38.07 and 41.10 percentage points. Deleting each eligible family once leaves dryer differences between 36.96 and 39.00 points and microwave differences between 38.17 and 42.94 points. The directions are positive in 76/81 and 29/33 eligible families. These deterministic deletion ranges identify concentration; they are not confidence intervals.

The pacifier material difference is only 0.23 points within families, with deletion range $[-0.94, 1.16]$. A pooled material association therefore gives little support for a universal material preference. Dryer noise also needs direction: the literal noise-only versus quiet-only comparison gives 7.89 points across 26 paired families, whereas suppressing local negation gives 13.29 across 24. The microwave equivalent has only one family. A merged noise topic near zero can hide opposite expressions; it cannot establish indifference to noise.

4.3 A fixed semantic audit and its action implications

Before reading review contents for this audit, we fixed 58 review–topic pairs across fourteen category/topic/low-star strata. Five reviews were selected per low/non-low stratum for dryer reliability, microwave reliability, dryer weight and dryer power/speed; three per stratum for microwave capacity and pacifier material/acceptance. Selection used the smallest SHA256 keys from a fixed seed and review ID within each stratum. Every selected headline, title and body was read, with no content-based exclusion. The stratum population and inclusion probability are retained.

One author AI coded the evaluated object, attribute direction, actual versus hypothetical failure, matched word's referent and use context. Ambiguous and mixed judgments remain visible. The table reports actual coded sample counts, not a claimed semantic accuracy or purchaser preference rate.

Table 6. Fixed audit: negative/positive/mixed/neutral/ambiguous counts.

Cat.	Topic	1–2 stars	3–5 stars
H	reliability	5/0/0/0/0	1/2/2/0/0
M	reliability	5/0/0/0/0	2/2/1/0/0
H	weight	3/2/0/0/0	2/0/2/1/0
H	power/speed	1/4/0/0/0	0/4/1/0/0
M	size/capacity	1/1/0/1/0	0/1/1/1/0
P	material	1/0/0/2/0	0/2/1/0/0
P	acceptance	1/0/0/2/0	0/2/0/0/1

Three findings change how a company should use the topic summaries. First, reliability hits often refer to an old machine rather than the reviewed one: two-star dryer R2H7DX2XUV3F8C says its *old* Remington died while criticizing the current model's airflow, heat and weight. Five-star microwave R24043XL4797KY praises a replacement while mentioning a failed GE. Reliability investigation should distinguish the current unit, an earlier unit of the same model and a different product.

Second, adverse overall stars can coexist with favorable attribute evaluations. One-star dryer R2OXGR21WD63IQ likes its lightweight construction but describes failed heat control; two-star R6TM8MUDWEYZC praises quick drying before stoppage. Four of five audited low-star power/speed reviews praise that attribute. Conversely, some favorable overall reviews still note burdensome weight. The data support testing comfortable handling and useful drying performance together with durability; they do not justify maximizing power or treating the topic's favorable pooled association as proof that all users prefer more of it.

Third, word senses and product scope create actionable misrouting. Pacifier acceptance hits include "take chances", "take to dry" and unpacking a box; material hits include packaging and a diaper bag's plastic hardware. Microwave R24810Q3ZRRTLZ mentions a *small kitchen fire*, not the new oven's small capacity. Such records belong in an audit queue, rather than an acceptance, material or capacity complaint count. These examples come from the fixed sample, and the complete coding retains both confirming and adverse cases.

The coding can be propagated as an explicit measurement scenario. For stratum s of size N_s with n_s audited records, estimate the coded count by $\widehat{T}_s = (N_s/n_s)\sum_{i\in s} z_i$. This expansion accounts for the deliberately unequal low-star sampling. Define Z as a reliability keyword hit coded as an experienced current-unit failure. Its expanded low-star contrast becomes 56.3 percentage points for hair dryer and 66.7 percentage points for microwave. This is a keyword-restricted coding scenario, not a corrected physical-failure rate. Only five records are coded per reliability stratum; programs agreeing on the arithmetic do not validate the AI interpretation.

For a stratum with x coded yes answers among n sampled hits, the finite count K is compatible with the observations whenever $x \le K \le N - n + x$. This bound needs no random sampling. For a separate conditional confidence calculation, assume the fixed SHA256 ranks approximate uniform simple random sampling without replacement and the same coding rule applies to every stratum member. Then

$$\Pr(X = x \mid K) = \frac{\binom{K}{x}\binom{N-K}{n-x}}{\binom{N}{n}}. \tag{3}$$

We retain integer K with both tails at least $0.05/(2\times4) = 0.00625$. Bonferroni gives at least 95% simultaneous coverage over the four reliability-hit strata under these assumptions. A deterministic chosen seed supplies no observed randomization guarantee; these are model-conditional intervals, with coding error still outside them.

Table 7. Reliability-hit strata: finite coded-count bounds.

Cat.	Stars	x/n of N	Compatible K	Conditional 95% K
H	1–2	4/5 of 492	[4, 491]	[98, 491]
H	3–5	2/5 of 467	[2, 464]	[13, 424]
M	1–2	5/5 of 204	[5, 204]	[76, 204]
M	3–5	2/5 of 51	[2, 48]	[2, 45]

For category low/non-low totals (L, H) and coded-flag totals (A, B), the contrast is $A/(A+B) - (L-A)/(L+H-A-B)$. It increases with A and decreases with B, so opposing interval endpoints give the following ranges without a new bootstrap.

Table 8. Low-star contrasts under declared coding and sampling (pp).

Cat.	Hit-flag compatible	Hit-flag conditional 95%	All-failure compatible
H	[-14.18, 89.14]	[4.55, 86.96]	[-99.77, 99.86]
M	[-25.93, 75.16]	[30.94, 75.16]	[-98.88, 99.61]

Both conditional hit-flag ranges remain positive but are broad; both fully compatible hit-flag ranges cross zero. More importantly, unhit reliability records were never audited. Allowing their current failures to range from zero to all 1,101/8,975 low/non-low dryer non-hits and 301/911 microwave non-hits makes the all-failure range nearly $[-100, 100]$ points. Thus the conditional hit result supports source reading within the lexical queue; it cannot certify the ranking of total failures or a component defect incidence.

A label-free threshold makes the investigation's vulnerability more concrete. Starting from (a, b, c, d), move t falsely flagged low-star reviews into the non-mention group. The contrast becomes

$$D(t) = \frac{a-t}{a+b-t} - \frac{c+t}{c+d+t}, \qquad t_* = \frac{ad-bc}{b+d}. \tag{4}$$

The zero threshold t_* follows by cross multiplication. It is a one-sided pooled misclassification scenario: it preserves the full cohort and does not claim every false hit is low-star or evenly spread across families.

Table 9. A one-sided keyword misattribution challenge.

Category	Initial pooled D (pp)	25% reassigned (pp)	Zero threshold
Hair dryer	40.38	32.14	84.0% of low hits
Microwave	55.17	47.13	86.9% of low hits

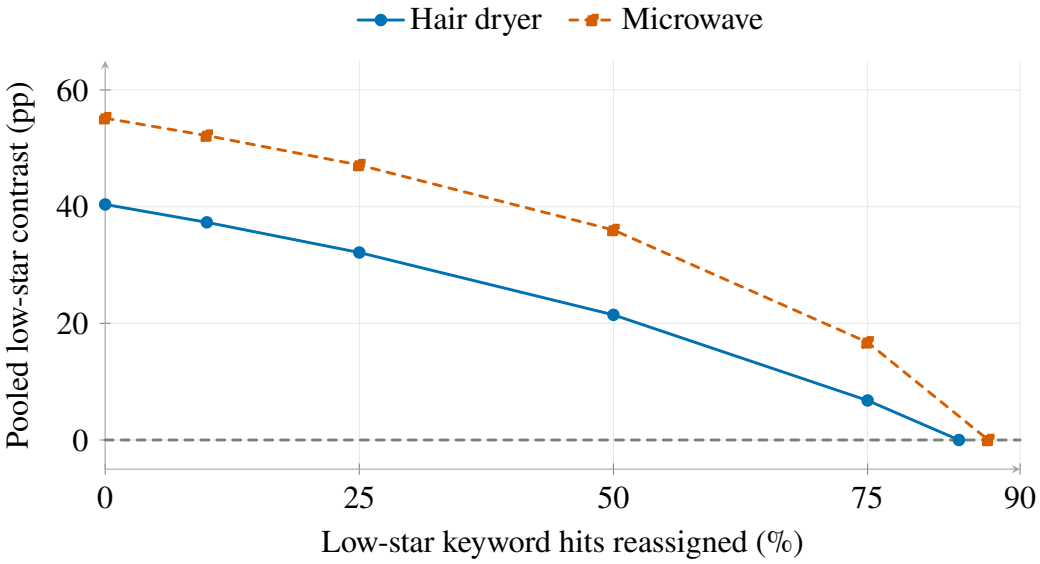


Figure 2. Reliability signals under misattribution.

Only false-positive low-star keyword hits are moved into the non-mention group; all target-cohort records remain. The marked zero is the exact scenario threshold, and connecting segments guide comparison.

The electrical reliability priority survives moderate amounts of this particular contamination, but it is conditional on the prevalence and distribution of misattribution. A company should first read a stratified set of its own flagged reviews and check the unit and failure event. If misattribution approaches the stated threshold or concentrates in key families, the lexical ranking must be recalculated before choosing a component investigation.

5 Monitoring reputation on a real calendar

5.1 A dashboard with a decision rule

At cutoff t, use reviews in $[t-90, t)$ calendar days. Display n, the five counts, q, mean stars, verified strata, current-unit complaint counts, vote coverage and conditional helpfulness. Record product variant and time since listing launch. Recent counts describe the current feedback window; cumulative averages provide context.

Let p_0 be the disclosed category pre-2014 low-star reference for the chosen population and $s = 10$ pseudo-reviews. The launch-policy example below uses the target-title cohort, whose microwave $p_0 = 0.441489$ differs from the full-file forecast training reference 0.537500, which pools future outcomes across eligible training windows rather than averaging all pre-2014 reviews; neither is the whole-file all-date low-star share 0.318266. Multiplying a binomial likelihood by the Beta prior gives

$$p \mid k, n \sim \text{Beta}(k + sp_0, n - k + s(1 - p_0)), \qquad E[p \mid k, n] = \frac{k + sp_0}{n + s}. \tag{5}$$

The 2.5% and 97.5% Beta quantiles form a model-conditional 95% interval [3]. For an illustrative *relative deterioration* alert, require at least ten reviews and $\Pr(p > p_0 + 0.10 \mid k, n) \geq 0.95$. Repeat it with $s = 2$ and 30; a policy that changes is prior-sensitive. The ten-point margin is an explicit management choice.

In the target-title last window, June 3–August 31, 2015 inclusive, 30 dryer, three microwave and 35 pacifier families have at least ten reviews. None crosses the relative rule at any of the three prior strengths. Yet the microwave reference itself is 44.1% low-star reviewing. A relative alert and an acceptable own-product risk are different decisions.

For acceptable reviewer-risk tolerance u, let $P = \Pr(p > u \mid k, n)$, false-investigation cost A, and missed-excess cost D. Investigating has expected loss $A(1 - P)$; deferring has DP. Investigate when $P > A/(A+D)$. The example family 565072108 has five low stars among eleven recent reviews. With $p_0 = 0.441489$ and $s = 10$, its posterior is Beta(9.414894, 11.585106). The relative-alert probability is 0.19492. Under an illustrative $u = 0.25, A = 1, D = 4$ policy, however, $P = 0.97333 > 0.20$, and losses are 0.02667 for investigation versus 3.89331 for deferral. This calculation demonstrates a budget trigger; its costs and tolerance are scenarios, not observed Sunshine figures.

A named analyst opens the five low-star records, confirms the variant and event, and routes repeated current-unit failure descriptions to engineering. An investigation closes on evidence from that work and the next fixed-window review, rather than on a recovered cumulative mean. A new listing with fewer than five recent reviews remains an insufficient-history case; specific credible incidents still deserve examination.

5.2 Time trends and listing mix

Falling low-star share suggests improving reviewer reputation, while repeated adverse issue descriptions can suggest deterioration. Review volume alone has several explanations, including sales and willingness to review. To expose composition changes, compare established families

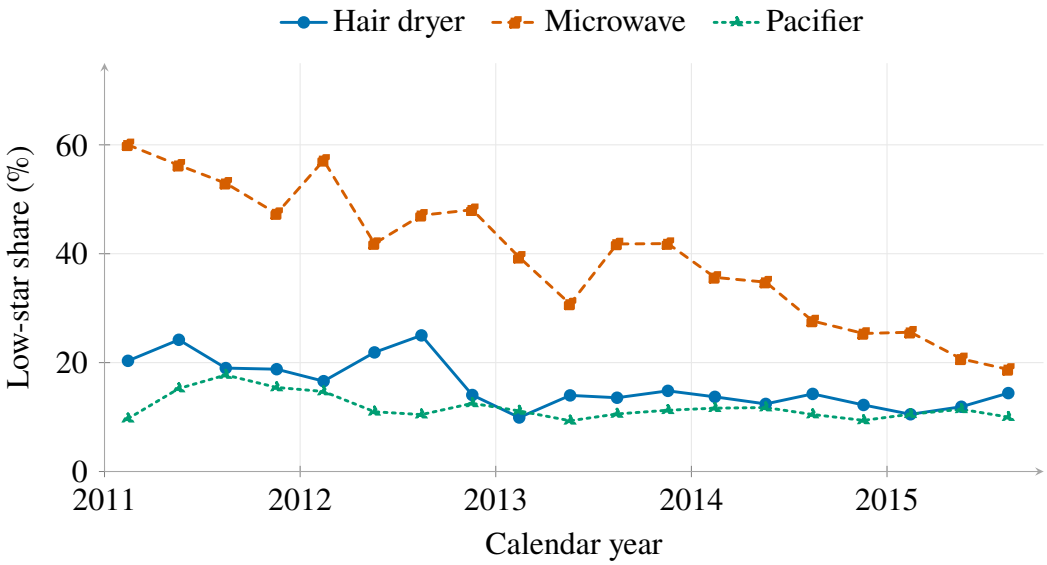


Figure 3. Calendar reputation profiles.

Points are calendar quarters with at least ten reviews; 2015 Q3 ends on August 31. Listing composition can differ between points.

with at least twenty reviews in 2014 and in January–August 2015, fixing both periods' weights to 2014 counts:

$$Q_\ell = \frac{\sum_j n_{j,2014} q_{j,\ell}}{\sum_j n_{j,2014}}. \tag{6}$$

Table 10. Target-cohort composition changes.

Category	Pooled 2014	Pooled 2015	Families	Fixed 2014	Fixed 2015
Hair dryer	13.0%	11.7%	28	12.0%	11.0%
Microwave	33.4%	24.1%	2	16.8%	16.0%
Pacifier	8.2%	9.2%	30	6.6%	9.3%

The target microwave pooled risk falls from 33.4% to 24.1%, but its two matched families change only from 16.8% to 16.0%. Pacifier pooled risk rises from 8.2% to 9.2%, while the fixed-family result rises from 6.6% to 9.3%. Both are reasons to track Sunshine's own listing and variant. The comparison covers established families and unequal seasonal periods, so it is descriptive rather than a causal annual-quality estimate.

6 Warning models and the information they require

6.1 A common prediction task

For question 2(c), define favorable future reputation as a low probability of one–two-star reviewing in the next ninety days. At January/April/July/October cutoffs from 2010 through April 2015, predictors use the preceding ninety days. Training ends in 2013; 2014 chooses the regularization; January and April 2015 provide complete future windows. The main comparison

requires at least five reviews on both sides, with 3,901 future reviews in 222 test windows from 138 families. A supplementary evaluation scores all 298 positive future windows eligible from their past, covering 4,119 reviews; two additional past-eligible windows have no observed future rating outcome.

Ratings use q, $(\bar{r} - 3)/2$, $\log(1 + n)/5$ and category indicators. Text replaces the rating terms with positive, adverse and defect/return mention fractions; combined includes both sets. No vote snapshot enters. For future low count y out of m reviews, logistic probability $p = (1 + \exp[-a - x^T b])^{-1}$ is fitted by

$$\min_{a,b} - \sum [y \log p + (m - y) \log(1 - p)] + \frac{\|b\|^2}{2C}. \tag{7}$$

Each window is represented by weighted positive and negative observations, reproducing the aggregate binomial likelihood [2]. Intercept a is unpenalized. $C \in \{0.1, 1, 10\}$ is selected by 2014 loss; the original candidates all select 0.1. No test refit occurs.

Review-weighted log loss $L = -\sum [y \log p + (m - y) \log(1 - p)] / \sum m$ evaluates the probability forecast. Brier loss replaces the numerator by $\sum [y(1 - p)^2 + (m - y)p^2]$. Large reviewed families consequently contribute more than sparse ones; an equal-weight listing objective would be different.

Table 11. Original snapshot-conditional probability scores.

Model	2014 loss	2015 loss	2015 Brier
Category	0.34352	0.34850	0.10000
Recent	0.33200	0.33129	0.09140
Ratings	0.32722	0.32402	0.09008
Text	0.33223	0.32969	0.09217
Combined	0.32685	0.32368	0.09012

Under the snapshot representation, combined log loss is 0.323683 versus 0.324025 for ratings, a gain of only 0.000342 nats per review; ratings have slightly lower Brier loss. The saved family-cluster interval for this gain is $[-0.000597, 0.001410]$, spanning zero. On all positive future windows, losses are 0.331041 and 0.331539. Sparse-future windows are harder to score than the primary reviewer-rich population. The target-title secondary comparison retains the same conclusion, with combined 0.32308 and ratings 0.32352.

Category differences also matter: original combined loss is 0.32636 for the dryer file, 0.47922 for the microwave file and 0.29777 for the pacifier file. These full-file forecast and subsequent flow comparisons retain surrounding products; they are file-level benchmarks, not verified forecasts for a pure pacifier cohort or Sunshine's new product. Microwave evaluation contains only eight families and 259 future reviews. Calibration bins expose uneven support: two dryer upper-risk points contain only 43 and 18 reviews, while a main bin contains 1,706. These results support a simple rating baseline as a candidate warning score, with text used to route complaints, rather than a promised predictive improvement from adding text.

6.2 The timestamp challenge changes the interpretation

The fixed semantic audit uncovered a concrete information failure. Review R1L4FFXO TTKWYT has a recorded date of December 2, 2006, yet its body includes updates dated September 26, 2009 and June 30, 2016. The latter is later than every recorded review date in the supplied files. The field's date therefore cannot certify that the complete current text existed in an earlier predictor window. Review R24810Q3ZRRTLZ explicitly changes its original four stars to one, showing that current rating values can also differ from their earlier versions.

A mechanical full-file scan records update/updated/edit/edited word markers and numeric dates in the next eighty characters. It finds 348 marker occurrences in 284 reviews, including 21 numeric-date candidates in 12 reviews and 5 occurrences after August 2015 in 3 reviews. These are candidate counts, not an unbiased estimate of real edits: a marker can concern a product update or quoted text, an undated edit remains temporally unresolved, and unmarked edits can be missed. Explicit examples establish the failure; the scan defines a reproducible sensitivity population.

We refit two deliberately conservative text-channel challenges with the same rating outcomes, calendar windows, training prefix, C choices and 2014 selection. *Prefix* tokenizes only the text before the first marker. *Blank-marked* assigns zero text flags to every marked review. It retains that review's rating and denominator, so the comparison does not gain by deleting inconvenient outcomes. Ratings are fitted on the same windows.

Table 12. Same-window text-channel sensitivity.

Measurement	2015 primary loss	All-positive loss	Primary Brier
Ratings	0.324025	0.331539	0.090079
Original combined	0.323683	0.331041	0.090121
Prefix combined	0.323654	0.331014	0.090115
Blank-marked combined	0.323684	0.331038	0.090128

The nominal combined advantage remains approximately 0.00034–0.00037 nats per primary-test review. Marker-bearing reviews occur in 23 of the 300 past-eligible 2015 windows (26 past-review occurrences). Removing their post-marker text or all their text barely changes the overall comparison. This weakens the explanation that explicitly marked tails alone created the nominal increment, while leaving its small magnitude and historical-availability uncertainty intact. These comparisons answer how much the nominal text increment depends on marker-bearing snapshot passages. They cannot reconstruct the historical text or stars of unmarked records. The defensible deployment plan is therefore to begin a timestamped own-product ledger containing the rating/text version actually received at each update, then validate a fixed ninety-day warning rule prospectively. The original numerical gains remain useful conditional benchmarks; their calendar split alone no longer warrants a strict historical-information claim.

7 Specific stars, sequences and subsequent reviews

A low-rating history can mean more one-star failures or more moderately dissatisfied two-star reviews. It can also end in a run of complaints or end in recovery. We distinguish these histories instead of compressing all of them into q.

For every window with at least five prior reviews, retain the next ninety-day count m, including zero. Families need at least three eligible windows. Define q_s as the fraction with s stars and use five stars as the omitted category:

$$\log(1 + m_{jt}) = \alpha_j + \tau_t + \gamma \log(1 + n_{jt}) + \sum_{s=1}^{4} \beta_s q_{s,jt} + e_{jt}. \tag{8}$$

Family effects adjust persistent differences, cutoff effects common calendar shocks and prior count review-flow persistence. Each β_s compares replacing five-star share with that particular star level, holding the other shares fixed. A ten-point substitution corresponds to $100[\exp(0.1\beta_s) - 1]$ percent in fitted $1 + m$, an association scale rather than a purchase effect.

Table 13. Specific-star shares and later review flow.

Cat.	Contrast	β_s	95% interval	10 pp (%)
H	1 vs 5	-0.286	[-0.940, 0.368]	-2.82%
H	2 vs 5	-0.109	[-0.781, 0.563]	-1.09%
H	3 vs 5	-0.086	[-0.663, 0.492]	-0.85%
H	4 vs 5	-0.213	[-0.839, 0.413]	-2.11%
M	1 vs 5	-0.498	[-4.184, 3.189]	-4.85%
M	2 vs 5	0.376	[-1.092, 1.844]	+3.83%
M	3 vs 5	-0.253	[-2.867, 2.362]	-2.50%
M	4 vs 5	-0.412	[-2.207, 1.383]	-4.04%
P	1 vs 5	-0.131	[-1.120, 0.858]	-1.30%
P	2 vs 5	-0.159	[-1.019, 0.702]	-1.58%
P	3 vs 5	-0.111	[-0.722, 0.501]	-1.10%
P	4 vs 5	0.008	[-0.537, 0.553]	+0.08%

Intervals use family-cluster CRV1 covariance with a small-sample correction and a t_{G-1} reference [4]. The twelve star-level coefficients form one exploratory Holm comparison family; the entire set is retained. Every individual interval spans zero and every Holm-adjusted value is 1.00. The joint four-star tests also provide no clear evidence of association (dryer $p = 0.910$, microwave $p = 0.892$, pacifier $p = 0.988$). The microwave one-star and two-star point estimates have opposite signs, illustrating why merging them loses information; their broad intervals prevent a confident differential recommendation. The full-star model covers exactly the same 461/50/353 windows and 63/8/58 families as the earlier low-star-share benchmark. Thus differences arise from the rating representation, not from changing the sample. Few microwave families warrant particularly cautious inference.

We also test histories whose latest three prior reviews have unambiguous day order. If any of those dates contains more than one review, sequence order cannot be reconstructed and

that window is excluded. After that exclusion, we recompute each family's eligible-window count and again require at least three qualifying windows; this yields the stated sequence population. Indicators distinguish three consecutive one-star ratings, three consecutive one–two-star ratings and three consecutive five-star ratings; the full rating histogram remains in the model. A sequence indicator with fewer than five eligible positive windows is not fitted.

Table 14. Latest-three-review sequence comparisons.

Cat.	Sequence	Windows	β	95% cluster interval
H	Three 1 stars	2	–	Too sparse
H	Three 1–2 stars	7	-0.063	[-0.563, 0.437]
H	Three 5 stars	75	0.056	[-0.110, 0.223]
M	Three 1 stars	2	–	Too sparse
M	Three 1–2 stars	4	–	Too sparse
M	Three 5 stars	10	-0.050	[-0.568, 0.467]
P	Three 1 stars	0	–	Too sparse
P	Three 1–2 stars	1	–	Too sparse
P	Three 5 stars	106	0.031	[-0.138, 0.200]

Sequence models use 348 windows/55 families for hair dryer, 42 windows/7 families for microwave, 266 windows/51 families for pacifier. All fitted sequence intervals cross zero. Triple one-star histories occur in only two dryer windows, two microwave windows and no pacifier windows; none supports a stable separate coefficient. The exact-string low-rating example is therefore rare in the estimable history, not a phenomenon proven absent from shoppers' experience. The stricter sequence population and its excluded ties are shown, so it is not disguised as the same full panel. Histories known only through current snapshot values still carry the rating-edit limitation. Sunshine can monitor concrete runs as review-reading prompts; these associations do not establish that shoppers saw a particular run.

Causal incitement cannot be identified from the supplied fields. Even if expected count is $E[m] = S\pi$, unobserved sales S and review propensity π can trade off. A count increasing from one to two fits sales increasing from ten to twenty at fixed $\pi = 0.1$, or unchanged sales of one hundred with π rising from 0.01 to 0.02. Both worlds can share the same observed stars and later reviews. Display exposure creates a further missing link. The company should test an incitement campaign only with separately collected impressions, displayed rating histories, purchases and review opportunities, using a defensible assignment design.

8 A practical launch strategy and its limits

Table 15 converts source reading into a company action. It uses the same fixed audit records: dryer R6TM8MUDWEYZC and R2OXGR21WD63IQ, microwave R2HKXN3 3E8UNK6 and R24043XL4797KY, and pacifier R1KAF3VO6292E6 and R10WJSO2V19VAJ. None is a new sample or a diagnosed returned unit.

Table 15. Object, event and action routes for the three launches.

Product	Object and event	Symptom or attribute	Action and review condition
Dryer	Current unit dries quickly, then stops; another current unit has failed heat control.	Speed is praised. Stoppage and heat control are symptoms; motor failure is unconfirmed.	Investigate repeated current events and inspect returns. Monitor handling/speed; recompute ranks after object corrections.
Microwave	Current unit runs without heat after 17 months; failed GE in another record is the prior oven.	No-heat alone names no faulty component. Prior-unit failure concerns a different object.	Investigate repeated current no-heat events. Monitor replacement experience; recompute counts after removing prior-unit mentions.
Pacifier	Missing plastic encasing on a novelty pacifier; rubber smell despite infant accepting the current pacifier; another brand of the same material is preferred.	Packaging cleanliness, nipple material, smell and infant acceptance are separate attributes.	Investigate specific packaging/odor complaints. Monitor age/use acceptance; remove packaging from material counts and recompute.

Use the absolute-risk loss trigger in Section 5 to open the review queue, then this object/event route to choose the inspection. Neither an alert nor a keyword assigns a failed component. New listings with too few reviews retain an insufficient-history status, while an individually credible event can open an investigation. Recompute rankings whenever source correction materially changes counts or family concentration; keep monitoring when the event is a prior-product comparison or an attribute preference without current failure.

Our strongest recommendation is a current-unit reliability investigation for hair dryers and microwaves. Large lexical associations persist within many families and under each family deletion. The fixed audit explains why reading the object and time is necessary, and the contamination threshold specifies a condition under which the lexical priority would have to be reconsidered. Engineering should record casing/handle, control and heating symptoms, check actual returned units, and only then identify the responsible component. The all-failure uncertainty range prevents interpreting the audit as a measured physical-failure ranking.

For dryers, assess comfortable handling and appropriate drying/heat control together with durability. Praise for speed can coexist with failure, while high power can create tangling or excessive heat in a different use group. For microwaves, give usable dimensions and plate fit and verify early-failure complaints; compactness suits some kitchens and boats, but compatibility and controls can dominate their overall evaluation. For pacifiers, separate true product acceptance and fit from accessories, novelty uses, packaging and other senses of “take”. A material choice requires an intended-use comparison; the aggregate material topic does not supply it.

After launch, assign an analyst to a weekly ninety-day dashboard and maintain timestamped versions from day one. Show all five stars, low-star risk/count/interval, verified strata, actual current-unit complaint counts and helpfulness reception. Management sets acceptable reviewer risk and an investigation budget separately from competitor averages. Text routes the investigation; ratings provide the simpler warning baseline to validate. A complaint investigation and an incitement experiment have different evidence requirements and should be managed as different decisions.

Independent CSV/calendar counts, rowwise forecast scoring, joint residualization and exact rational reclassification thresholds check the calculations used in these answers. They leave semantic truth, reviewer selection and missing historical versions unresolved. Purchaser defect incidence, sales/profit response and the causal benefit of changing a feature require additional observations. This division preserves concrete actions: investigate the supported electrical reliability themes, avoid universal pacifier material claims, monitor each own listing's dissatisfied tail, and validate warning rules against a ledger with known information times.

References

[1] COMAP. *2020 MCM Weekend 2, Problem C: A Wealth of Data*. Original three-page problem, glossary and supplied `hair_dryer.tsv`, `microwave.tsv`, `pacifier.tsv`. All product evidence comes from these files.

[2] scikit-learn developers. *LogisticRegression*, version 1.9.1 documentation. https://scikit-learn.org/stable/modules/generated/sklearn.linear_model.LogisticRegression.html. Consulted October 13, 2026; weighted likelihood and penalty interface.

[3] SciPy developers. *scipy.stats.beta*. https://docs.scipy.org/doc/scipy/reference/generated/scipy.stats.beta.html. Local version 1.18.1; quantiles and tails for the explicitly derived model.

[4] Statsmodels developers. *RegressionResults.get_robustcov_results*. Version 0.15.0. https://www.statsmodels.org/stable/generated/statsmodels.regression.linear_model.RegressionResults.get_robustcov_results.html. Consulted October 13, 2026; CRV1 small-sample and group-degrees-of-freedom options.

A Reproduction files and fitted parameters

This paper accompanies the case folder `strong-o-2020-c`. Its `reproducibility-index.json` identifies the input files, fixed sample, complete AI codes, original result snapshots, new model/validator, current run, saved predictions, panel matrices, figure arrays and this paper generator by relative path and SHA256. `REPRODUCTION.md` gives executable commands. The complete model parameters, transformations, C candidates and selection losses are available in `model-parameters.json`; it includes the original full and target-cohort rating/text/combined candidates and each new censoring candidate. These are files supplied with the paper, not an unspecified external archive.

The recommended original ratings baseline uses risk k/n, scaled mean $(\bar{r}-3)/2$, log count $\log(1+n)/5$, and microwave/pacifier indicators with dryer as baseline. Its selected $C = 0.1$, aggregate likelihood and training prefix are those in Section 6. The full numerical coefficients are shown below; current field availability must be established before deployment.

Table 16. Recommended ratings-baseline parameters.

Term	Coefficient
Intercept	-1.143044995
risk	0.334748042
mean_scaled	-0.906890315
logn	-0.394556610
microwave	0.675750434
pacifier	-0.517704907

Report on Use of AI

This is an AI-authored historical-practice learning revision using example team number 7391856, with no contest submission or award claimed. The author Codex AI read the original problem and preceding own paper, fixed and read the 58-review sample, supplied the conditional semantic codes, wrote new numerical and independent arithmetic checks, interpreted the results, wrote the paper and company letter, and generated and inspected the actual PDF. Arithmetic checks use different computations with common input data and definitions; they do not independently certify the AI's semantic interpretation. No human semantic gold labels or independent human mathematical work are claimed.

The project had previously read two same-topic papers after an earlier autonomous seal. That exposure remains part of this learning case. The present author read only its own task/data/code/current evidence and generic resources; no new same-topic reference paper or outside student script was used. The current run and evaluation are post-test secondary comparisons, not a new first-autonomous result or pristine prediction test. An author-external initial review prompted conditional finite-population audit ranges and source-bound action routes. The recovery author reused the complete run5 results without refitting forecasts, recorded a failed validation attempt and its repair, and added an independent exact-combinatorial endpoint check. The coordinating AI's later whole-paper review belongs to its separately bound review record.

The runtime identifies the assistant as Codex based on GPT-6. An exact provider model identifier, actual model tokens and costs, and a complete provider chat export are unavailable. Retained argv, source snapshots, failures, timings and hashes are technical evidence rather than a complete conversation log. The accompanying records preserve the registered total budget, new calculation stages and prior historical identities. A practice AI report is supplied separately from the 2020 solution-page boundary; it does not claim that a current AI policy applied to the original contest year.